Project Proposal

Title: College Test, Assessment & Student Performance Management System

Submitted to: Department of Computer Science

The Islamia University of Bahawalpur (IUB)

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1. Introduction and Problem Statement

Colleges conduct weekly and periodic tests, but most still manage schedules, papers, marks and parent communication manually or in scattered Excel files. This causes slow result preparation, data entry errors, loss of historical records, and no clear view of student progress for the Principal. This project proposes a centralized web application that manages the complete testing lifecycle of a college in one place.

2. Objectives

- Digitize academic setup: sessions, classes, sections, groups, subjects and teachers.
- Plan weekly test schedules (Test Groups) and individual tests with papers and scanned copies uploaded, even from a mobile phone.
- Enable fast marks entry (manual and Excel import) with validation, pass/fail calculation, absent status and result finalization with locking.
- Generate test, weekly, class, subject and individual student reports (Print, PDF, Excel).
- Prepare weekly performance messages for parents with preview and Principal confirmation via WhatsApp Web.

3. Scope and Key Features

The system supports three roles (Admin, Principal, Clerk) with session-based role-based access control. Main modules: Dashboard with live statistics, Students, Classes/Groups/Subjects, Teachers and assignments, Test Groups and Tests, Results, Reports, WhatsApp Reports, Users and Permissions, Audit Logs and Recycle Bin. All deletions are soft deletes, sensitive changes (especially marks) are audit-logged, and unlocking finalized results is recorded.

4. Proposed Technology

Layer	Technology
Backend	Core PHP 8.x (modular MVC-style)
Database	MySQL 8.x (normalized, indexed)

Layer	Technology
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript, jQuery, DataTables
Communication	AJAX with consistent JSON responses

5. Methodology

The project will follow an iterative development model: requirements analysis (SRS v1.0 completed), database and system design, module-wise implementation, testing, and deployment. Server-side validation, RBAC on every endpoint and server-side DataTables will be applied throughout for security and performance.

6. Expected Outcomes

A responsive, easy-to-use system that a clerk can operate and that gives the Principal accurate, up-to-date insight into class, subject and student performance, while keeping complete academic history safe. The architecture also allows future expansion, such as attendance, student/parent portals, online tests and the official WhatsApp API.

7. Tentative Timeline

Analysis and design (3 weeks), core modules (5 weeks), tests and results (4 weeks), reports and WhatsApp (3 weeks), testing and documentation (3 weeks).